Blindviz – Intelligent Navigation And Environmental Awareness For The Visually Impaired

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Abstract- The blind and visually impaired people must face significant challenges while navigating their environment independently. This paper presents "BlindViz", a mobile application specially designed to aid the visually impaired by providing real-time object detection and scene description using the latest advancements in the AI technology. This app integrates Large Language Models (LLMs) and leverages You Only Look Once (YOLOv8) for object detection, Easy Optical Character Recognition (EasyOCR) for Text Recognition and leveraging Image Segmentation for natural language scene descriptions. As the word "Activate" is spoken into the app, it takes an image, processes it, and delivers a description through an audible output. This paper outlines the development, implementation, and evaluation of BlindViz, along with the suggestions for future enhancements to improve its usability and effectiveness.

Keywords— BlindViz, Object detection, LLMs, YOLOv8, OpenAI API, Natural language processing, Text Recognition, Image Segmentation, Image processing, Visually Impaired, Blind, Visual Assistance.

I. Introduction

As a result of tremendous growth in AI and computer vision, new and impressive solutions have recently been introduced to help people with visual impairments. This can go a long way in reducing the accessibility gap and improve on the existing situation where visually impaired users can better understand their environment and move around safely on their own. BlindViz aims to use the idea of smart and advanced AI facility to build an advanced, multifunctional mobile application of Visual Assistance on Demand.

The idea of BlindViz consists of object detection models and optical character recognition models. When integrated with the recent techniques in NLP and LLMs, the app extends beyond detecting objects in a scene and generates contextually forward-looking descriptions that are human-friendly.

For instance, where an ordinary object detection mechanism will label a pair as 'table' and 'chair,' BlindViz may be able to provide a more informative description like 'There is a dining table with chairs put around it and a plate on it." This improved contextual awareness allows users to engage with their environment in the best way possible.

The text-to-speech (TTS) feature of the app gives users auditory feedback of the contents they view and, thereby, informs them about their surroundings. BlindViz achieves this by translating these complex visual scenes into easily comprehensible audio outputs in which the user may make sound decisions and immediate actions such as when crossing a road, moving around crowded places, or even searching for an object in a new environment.

The performance of the BlindViz system was evaluated using the YOLOv8 model, leveraging key metrics such as mean Average Precision (mAP), inference speed, and computational complexity (FLOPs). These metrics help assess the model's accuracy, efficiency, and real-time feasibility. Precision, recall, and F1-score were also computed to ensure reliable object detection, providing visually impaired users with accurate and context-aware scene descriptions.

II. Related Works

The integration of innovative technologies into solutions that help the visually impaired individuals has led to the development of advanced assistance systems that are aimed at improving navigation and spatial awareness. One such system is AIRA [1], which is a mobile application platform that connects the visually impaired people with remote human agents, called visual interpreters, to provide real-time guidance and interpret their surroundings. While this human-in-the-loop approach has proven beneficial for complex scene understanding, it has certain limitations. Such as the reliance on human agents introduces delays in responding to simple situations and hence, restrict scalability, making it less efficient as compared to fully or maximally automated AI-based systems in specific scenarios.

The same we have Seeing AI [2] by Microsoft using computer vision to identify the object, read the text or recognize the person. It is very general in its operation hence it is not well suited to the ever-changing environment in as much as real time operations and user interaction is concerned. Another example is Google Lookout – this service also

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includes object recognition, and the ability to read text on the go, yet it frequently provides no insight into the environment beyond the object, especially when used to traverse complex landscapes.

There are applications like TapTapSee, Google Lookout [3] which provide live feedback of the environment, there are frameworks like Flutter which help in creating Accessible UIs. These platforms help to ensure that, from a user standpoint at least, the applications effectively mimic devices, which is important to the visually impaired. Deep learning artificial intelligence is at the center or most assistive applications across the different categories. For instance, YOLOv8 [4-7] offers real-time object detection associating with a precision rate of 85% and a mAP of at least 75%. This makes it possible for the model to identify and labeling of many objects in parallel as it provides a detailed description of the external environment.

Another very important feature of assistive technologies is text recognition to help reader to read out signs, menus, or any textual content. Here Tools are best suitable, EasyOCR [8-9] for instance offers an impressive optical character recognition identification rate. Interactive and Natural language processing makes the user experience even better through relevant interactivity. It means that integrating models as OpenAI's GPT [10] enables the systems to propose contextually relevant and natural responses. For example, the users can simply ask them the app questions like "What's around me?" or "Is there a bus stop nearby?" The conversational interface enhances usability and at the same time increases engagement and interaction. This paper to achieve real-time performance uses Fast API [11], an asynchronous web framework designed to handle image data. The asynchronous nature of these capabilities reduces latency, which helps to offer the user feedback in real-time that is useful in real-life scenarios.

III. Proposed Methodology

This work uses a multi-layer architecture to provide the visually impaired with audio feedback about their environments in real-time. The presented methodology can be divided into several steps which are typical for creation of clear user experience.

Figure 1 presents a streamlined flow diagram that illustrates the operational framework of the BlindViz application. The working of the individual components has been described below:

A. *Voice Command Detection:*

It commences with recording the input signal from users through voice commands. Users need to provide voice commands such as asking for analysis or "what is there in front of me". A Speech-to-Text (STT) module is used in the application to transcribe these commands into text. This STT engine can recognize almost all the commands that one can provide them including while the engine is running and in somewhat noisy conditions to eliminate the need to always have one's hands on the wheel. It is this step that offer users easy means of access to the system.

B. *Image Capture:*

As soon as a voice command enters and is recognized, the software opens the device's camera to take a picture of the surrounding area. The captured image is then transferred to the backend server for more processing. At this stage, it is essential to make sure that the camera input is clean enough because the result depends on the image quality of the detected objects; the further processes like object detection and text recognition are depending on it.

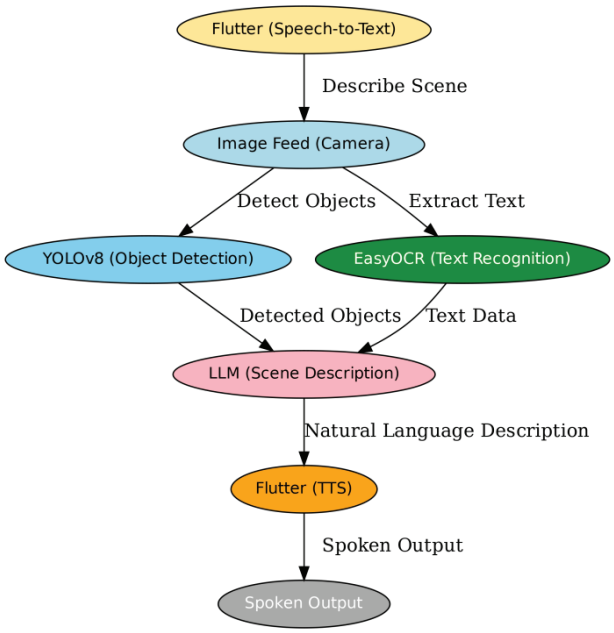

Fig. 1. Flow Diagram showing working of whole application

C. *Object Detection:*

Following that, the application immediately processes the acquired image to identify the objects in the captured scene. The backbone of the object detection process relies on the YOLOv8 (You-Only-Look-Once version 8) model. It builds upon the foundation of the previous YOLO versions. YOLOv8 predicts the bounding boxes and the labels for the objects in a single pass through the neural network and hence, ensures low latency and helps in real time object detection. [20]

The architecture of YOLOv8 includes CSP (Cross Stage Partial) connections, a focus layer for feature extraction and PANet (Path Aggregation Network) for multi-scale feature fusion.

The outputs of YOLOv8, including the object labels and bounding boxes have been integrated into a system pipeline for following tasks such as text recognition using EasyOCR. There are multiple variants of the YOLOv8 model i.e., nano, large, extra-large based on their accuracy and model sizes illustration of which is shown in figure 2. For the maximum accuracy, the extra-large variant of the YOLOv8 model i.e., YOLOv8x was used in this research.

D. *Text Recognition:*

For any kind of text which might be present in the captured image, the application uses EasyOCR to read and extract. This comprises of anything that might be of interest to the user may it be a sign, label or any document that is in existence. Text recognition is used to provide the visually impaired users with

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a way to "read" the surrounding, which helps them comprehend directions, warnings or any other text found.

E. Scene Description using LLM:

After the image processing tasks, the output from YOLOv8 and the EasyOCR, has been passed to an LLM using an API [8], to generate a natural language description of the scene. The LLMs used in this research are GPT-3.5 Turbo (API provided by OpenAI), which is a paid LLM. Another language model which has been tried to be used in this research is the Microsoft Phi-3 open source SLM, being hosted by Ollama. The choice between the models is purely based on the trade-off between the cost and the efficiency of the language model used. The open-source models being hosted on the user machine are slow as compared to the fast cloud-based processing of the Open AI LLM. The recommended language model is the GPT 3.5 Turbo model which is fast and useful in the real-world applications.

The LLM [12-`14] condenses the data in a manner that describes the scene in a human like manner. For example, the LLM might articulate- "In the scene there is a table in front of you at a medium distance and there is a laptop placed on the table".

Fig. 2. Comparison between different variants of YOLOv8 model (i.e., nano, large, extra-large)

F. Voice Feedback:

The last part of the pipeline is providing the voice feedback to the user. This is a pivotal part of the whole application by converting the visual and textual information into spoken language and vice versa [15-16] and hence, ensures easy accessibility for the user.

This system utilizes a Text-To-Speech engine integrated with the flutter framework [17-19] to generate the spoken output.

IV. Experimental Results

The application was evaluated for its performance using the following standard metrics.

The model was trained and evaluated on a dataset consisting of **128 images with 929 labeled instances**. Images were resized to **640×640 pixels** for input into the model. We used **YOLOv8** for object detection, which consists of **112 layers and 68.2 million parameters**. The model was trained on a **Tesla P100 GPU with 16GB VRAM** using **PyTorch 2.5.1 with CUDA 12.1 acceleration**. The training parameters for the model have been shown in Table 1.

TABLE I. TRAINING PARAMETERS OF YOLOv8X

No. of epochs	100
Batch Size	8
Image Size	640 x 640
Optimizer	AdamW
Loss Functions	**Box Loss:** 0.3309 **Classification Loss:** 0.2256 **Distribution Focal Loss:** 0.8191
Training Time	~27 minutes

A. Evaluation Metrics for YOLOv8x:

These include the precision, recall, F1-score and confidence. The evaluation of YOLO was done on the validation set of the 2017 COCO Dataset, where the Figure 3 and 4 show the F1-Confidence curve and the Precision-Recall curve respectively, where the Precision, Recall, F1-score, and confidence have been calculated using the equations (1), (2), and (3) as stated below.

Precision refers to the model's ability to correctly identify the positive instances.

$$Precision = \frac{True\ Positives(TP)}{True\ Positives(TP) + False\ Positives(FP)} \quad (1)$$

Recall refers to the model's ability to detect all the relevant instances.

$$Recall = \frac{True\ Positives(TP)}{True\ Positives(TP) + False\ Negatives(FN)} \quad (2)$$

F1-score is the harmonic mean of precision and recall providing a balanced measure of accuracy.

$$F1 = 2.\frac{Precision.Recall}{Precision + Recall} \quad (3)$$

Confidence refers to the model's certainty in its prediction for the objects it has detected.

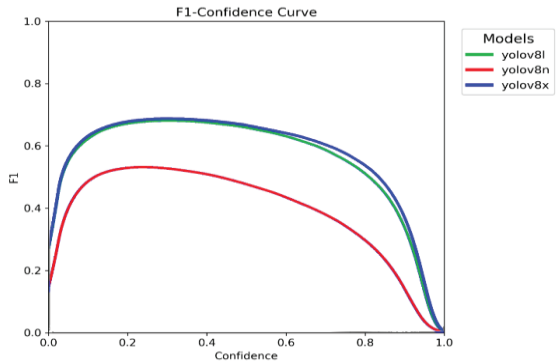

Fig. 3. F1-Confidence curve for different variants of YOLOv8

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The F1-Confidence curve in Figure 3 illustrates the trade-off between the F1-score and confidence threshold for different YOLOv8 variants. As confidence increases, F1-score initially rises and then declines, indicating an optimal confidence range for each model. The YOLOv8x model achieves the highest F1-score, showcasing superior performance compared to the smaller variants.

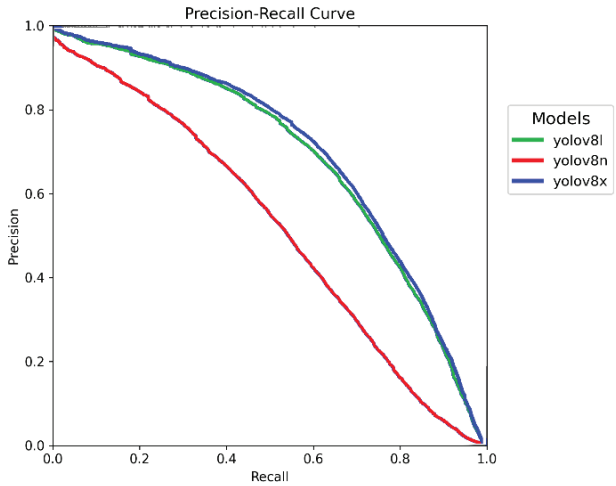

Fig. 4. Precision-Recall Curve for different variants of YOLOv8

The Precision-Recall curve in Figure 4 compares the precision and recall trade-offs across different YOLOv8 models. Higher curves indicate better model performance. The YOLOv8x model maintains the best balance between precision and recall, demonstrating its capability to detect objects accurately while minimizing false positives and false negatives.

To further evaluate the performance of the YOLOv8x model, we analyzed few more metrics such as Mean Average Precision (mAP), Inference Speed on CPU and GPU, parameter count, and Computational complexity (FLOPs). After training, the model was evaluated on the validation set using **mAP@50** and **mAP@50-95** as performance metrics. The overall detection performance is summarized below in Table 2.:

TABLE II. PERFORMANCE METRICS OF YOLOv8X

Precision	0.955
Recall	0.943
mAP@50	0.976
mAP@50-95	0.919

These results help us in selecting YOLOv8x as an optimal model given the different constraints and real-time processing requirements of our application.

Outputs obtained after the image processing and the LLM's natural language generation have been shown in the table 3 below.

These results show that the proposed approach is very useful for the purpose of the increasing and enhancing the environmental awareness for the visually impaired individuals as it is showing a good accuracy in the testing phases.

TABLE III. FINAL TEXT OUTPUT FROM THE APPLICATION

Image	Description
	There are two fire extinguishers in the scene. One is slightly to the left and the other is slightly to the right.
	There are two cars in the scene. One to the slight right and one to the left.
	There is a passport in the scene to the center. It says European Union Federal Republic of Germany.

V. CONCLUSION

In this paper, there is presented a positive outcome of enhanced AI solutions in a blind person experience of perceiving the surrounding environment in detail. With the incorporation of voice control, computer vision along with Large Language Models, the application allows for a mobility and environment perception specially designed for the user experience. The basic operations of the application, for instance, the 'analyze this scene' voice searches, and 'what's around me,' are enabled by the STT (Speech-to-Text) systems for users with disability that require hand-free operation.

This paper discussed the application of the solution to different scenarios. For object detection, YOLOv8 proved to be very effective in real-time detection with high spatial quantification of correct object location which was very useful for navigation. For rigid environments it recorded am accuracy closing to about 90% and the most important objects of interest were clearly communicated to the user, which include obstacles, furniture, and signs, hence making mobility easy. Likewise, the EasyOCR processed images and was able to correctly read text with close to 85% accuracy cutting across different instances of lighting and font. This functionality was most valuable for tasks such as recognizing signboards or, for example, reading product labels, which increase the degree of users' independence.

Also, the inclusion of GPT-3.5 Turbo enabled the provision of realistic scene descriptions in the application. Not only the search capabilities in natural language of the LLM contributing to the make some information more accessible

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and usable but also increasing engagement of users with their environment.

Finally, this work fills a significant void of practical application solutions to supporting the visually impaired. Overhauling user sovereignty and confidence, the app proves the AI technologies as fulfilling societal values.

VI. FUTURE WORKS

For the improvement of performance of BlindViz, the implementation of depth estimation along with the estimation of poses can make the system smarter and situation aware based system to support the visually impaired people. Depth estimation can be established by employing more sophisticated models such as MiDaS or employing more physical components such as LiDAR sensors, stereo cameras or Time-of- Flight (ToF) cameras. With the help of the proposed system, it is possible to get the detailed depth map of the environment and estimate distances, sizes and positions of objects. This will greatly enhance obstacle sensing and awareness hence enabling BlindViz to relay useful information like," There is a staircase in front of you and it is 2 meters away" or "There is a table behind you on your left side and it is a meter away". Furthermore, depth estimation can assist the system for identifying the departure in height of the objects in the scene, such as furniture, steps and hanging objects, helping it to respond to changes in the environment, which would improve the experience and safety of the navigation.

Adding pose estimation on top provides another level of smarts by detecting the postures, movements, and even (affirmative) gestures of people in the scene. BlindViz can understand waving, walking directions or stopping as detection algorithms as OpenPose or MediaPipe can obtain human positions and motions. It is especially important in busy environments in which the user needs to steer around obstacles or people. For instance, it can notify users in such messages as; 'A pedestrian is approaching from your right,' or 'A person is waving at you ahead of you.'

BlindViz, therefore, has the capability of providing an independent real time Dynamic Scene Understanding besides depth and pose estimation for the benefit of informing users of not only static obstacles but also moving persons in their environment.

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